

Lab: Derivation Exercise — Parameter Learning and Model Reduction

Objective

Derive the parameter learning equations and Bayesian Model Reduction formula, applying them to concrete examples.

Part 1: Dirichlet-Categorical Conjugacy

Goal: Derive the conjugate update from first principles.

Setup: Prior: $p(\theta) = \text{Dir}(\theta; \alpha_1, \alpha_2, \alpha_3)$ with $\alpha = [2, 3, 1]$. Likelihood: $p(x|\theta) = \text{Cat}(x; \theta)$. Data: 10 observations: $\{1, 1, 2, 1, 3, 2, 1, 1, 2, 1\}$ (category counts: $n_1=6, n_2=3, n_3=1$).

Task:

1. Write Bayes' rule: $p(\theta|x_{1:N}) \propto p(x_{1:N}|\theta) \cdot p(\theta)$
2. Expand the likelihood as $\prod_{i=1}^N \theta_{x_i}^{n_{x_i}}$ and the prior as $\prod_{j=1}^3 \theta_j^{\alpha_j-1}$
3. Show the posterior is $\text{Dir}(\theta; \alpha_1+n_1, \alpha_2+n_2, \alpha_3+n_3) = \text{Dir}(\theta; 8, 6, 2)$
4. Compute the posterior mean $E[\theta|\text{data}]$ and compare with the MLE

Write your response here

Part 2: A-Matrix Learning in a POMDP

Goal: Track how the A matrix evolves with experience.

Setup: A 2-state, 2-observation POMDP. Initial A matrix concentration:

$a = [[10, 1], [1, 10]]$ (state 1 $\rightarrow$ observation 1; state 2 $\rightarrow$ observation 2)

The agent experiences 20 trials where state 1 produces observation 1 (18 times) and observation 2 (2 times).

Task:

1. Apply the Dirichlet update rule: $a_{ij} \leftarrow a_{ij} + n_{ij}$
2. Compute the updated concentration parameters
3. Compute the updated expected A matrix (posterior mean)
4. How has the A matrix changed? How much more confident is the agent?

Write your response here

Part 3: Bayesian Model Reduction

Goal: Apply BMR to decide whether a parameter should be pruned.

Setup: After learning, a 3-state POMDP has posterior A matrix concentrations:

$a_{\text{full}} = [[50, 2, 1], [2, 48, 1], [1, 1, 45]]$

Consider a reduced model where states 2 and 3 are merged (reducing from 3 to 2 states), with reduced prior concentrations:

$\tilde{a}_0 = [[50, 3], [2, 49]]$

Original prior: $a_0 = [[10, 1, 1], [1, 10, 1], [1, 1, 10]]$

Task:

1. Compute the BMR evidence ratio using: $\Delta F \approx \ln B(\tilde{a}_0) - \ln B(a_0) - \ln B(\tilde{a}) + \ln B(a)$
2. Where $\tilde{a} = a + \tilde{a}_0 - a_0$
3. Determine whether the reduced model is preferred ($\Delta F < 0$)
4. Explain what this means: is the third state "doing work" or can it be pruned?

Write your response here

Part 4: Learning Rate and Forgetting

Goal: Analyze the effect of learning rate and forgetting on parameter trajectories.

Task:

1. For the Dirichlet update $a \leftarrow a + \eta \cdot n$, analyze the effect of $\eta < 1$ (slow learning) and $\eta > 1$ (fast learning)
2. Consider a forgetting mechanism: $a \leftarrow \lambda \cdot a + \eta \cdot n$, where $\lambda < 1$ is a decay factor. Show that this implements exponential forgetting
3. Compute the effective "memory window" as a function of λ
4. Discuss: when is forgetting adaptive? When is it maladaptive?

Write your response here

Part 5: Synthesis

In 200 words, explain how parameter learning (Dirichlet updates) and structure learning (BMR) together enable the agent to both refine and simplify its generative model, and how this maps onto the neuroscience of waking learning and sleep consolidation.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Bayesian updating	Dirichlet-Categorical conjugate derivation
2	Matrix learning	A-matrix concentration update tracking
3	Model comparison	Bayesian Model Reduction evidence ratio
4	Dynamical analysis	Learning rate and forgetting trade-offs
5	Conceptual integration	Connecting learning math to neuroscience